When Is the Mean Nearly Normal?

AP Statistics · Unit 2 · Topic 2.12 · TEACHER KEY

CED TETHER

- **Skill 3.C** — Describe probability distributions.
- **Enduring Understanding UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.H** — Estimate sampling distributions using simulation. [Skill 3.C]
- **EK UNC-3.H.1** — A sampling distribution of a statistic is the distribution of values for the statistic for all possible samples of a given size from a given population.
- **EK UNC-3.H.2** — The central limit theorem (CLT) states that when the sample size is sufficiently large, a sampling distribution of the mean of a random variable will be approximately normally distributed.
- **EK UNC-3.H.3** — The central limit theorem requires that the sample values are independent of each other and that n is sufficiently large.
- **EK UNC-3.H.4** — A randomization distribution is a collection of statistics generated by simulation assuming known values for the parameters.
- **EK UNC-3.H.5** — The sampling distribution of a statistic can be simulated by generating repeated random samples from a population.

Essential question: How do population shape and sample size determine whether a normal model is trustworthy for sample means?

DOK-3 skill: 4.B · **FRQ pattern:** `clt-when-does-normality-apply`

DOK rationale: Part (c) weighs population shape, independence, and sample size to judge two normality claims and distinguish a model for sample means from a model for individuals.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Does averaging four values erase a long right tail automatically?
- Which distribution does the CLT describe: commuters or sample means?
- What happens to $15/\sqrt{n}$ when n grows from 4 to 40?
- Which probability calculation becomes questionable if the n equals 4 shape is not normal?

[WARN] LOOK FOR

- Invoking the CLT for n equals 4 without considering the strong population skew.
- Saying forty individual commute times have a normal distribution.
- Using 15 minutes, rather than $15/\sqrt{n}$, for the spread of sample means.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Rejects automatic normality at n=4 using population right skew.

- **judgment-2** (required) — Supports n=40 as more defensible using independence and the CLT, without claiming a universal guarantee.
- **judgment-3** (required) — Distinguishes sample means from individual times and explains why a small-sample normal tail can be unreliable.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

Common mistakes

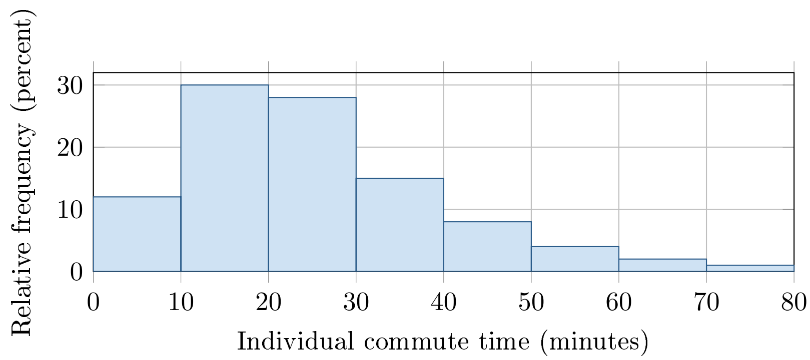
- Treating the central limit theorem as applying at every sample size
- Saying the individual commute-time population becomes normal when n increases
- Dividing the mean, rather than the standard deviation, by $\sqrt{n}$
- Using the number of repeated samples instead of the observations per sample as n

[STAR] THE PROBLEM — annotated

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: "Any average is approximately normal, even for $n = 4$."

Sol: "With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible."



FIRST TAKE — one sentence before you work the parts

Which would you expect to smooth out the long commute times more: averaging 4 people or 40? Give one reason.

Key: Not graded. Expect 'averages are normal' as an initial rule; the revision should add population shape and sufficiently large n.

[BOOK] CLT: SHAPE, SIZE, AND THE RIGHT DISTRIBUTION (keep on the page)

The sampling distribution of $\bar{x}$ contains the sample means from all possible random samples of one size n. For independent observations, $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$. The **central limit theorem** says that when n is sufficiently large, the sampling distribution of $\bar{x}$ is approximately normal even if the population is not. Stronger population skew generally requires a larger n; a small sample can retain that skew. The CLT concerns sample means, not individual observations.

DOK 1 **(a)** Identify the population shape and state the population mean and standard deviation with units.

Key: The population distribution of individual commute times is strongly right-skewed. Its mean is $\mu = 25$ minutes and its standard deviation is $\sigma = 15$ minutes.

DOK 2 **(b)** For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.

Key: For either sample size, $\mu_{\bar{x}} = \mu = 25$ minutes. For $n = 4$, $\sigma_{\bar{x}} = 15/\sqrt{4} = 7.5$ minutes. For $n = 40$, $\sigma_{\bar{x}} = 15/\sqrt{40} \approx 2.3717$ minutes, or about 2.37 minutes.

DOK 3 **(c)** In 3–4 sentences, decide whose claim is better supported. Use the population shape, independence, and the two sample sizes. Explain which distribution the CLT describes and why using a normal curve for averages of 4 could be unreliable.

Key: Sol's claim is better supported: a long right tail can remain when only four commute times are averaged. Independent samples of 40 make an approximately normal distribution of sample means more defensible by the CLT. The CLT describes sample means, not individual commute times. Using a normal curve for averages of 4 could give unreliable tail probabilities.